

MA 301
Test 3
Spring - 2014

Problem	points	out of
1		10
2		10
3		6
Score		26

Name: SOLUTIONS

HR: _____

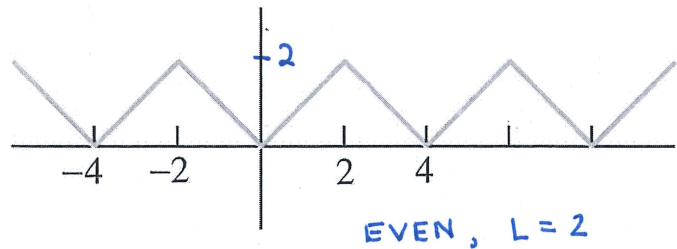
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Formula Sheet	
Fourier Series:	$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L} \quad \text{where} \quad d_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi n x/L} dx$
if $f(x)$ is even then	$d_n = \frac{1}{L} \int_0^L f(x) \cos\left(\frac{\pi n x}{L}\right) dx$
if $f(x)$ is odd then	$d_n = \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{\pi n x}{L}\right) dx$

Problem 1: Consider the sawtooth function, $f(x)$, below. Determine the Fourier Series of $f(x)$ in :

(a) Exponential Form: $e^{i\pi nx/L}$

(b) Cosine & Sine Form: $\cos\left(\frac{\pi nx}{L}\right)$ & $\sin\left(\frac{\pi nx}{L}\right)$



$$d_0 = \frac{1}{2} \int_0^2 f(x) dx = \frac{1}{2} (2) = 1$$

$$d_n = \frac{1}{2} \int_0^2 x \cos\left(\frac{\pi n x}{2}\right) dx$$

$$= \frac{1}{2} \left[\frac{2x}{\pi n} \sin\left(\frac{\pi n x}{2}\right) - \frac{2}{\pi n} \int \sin\left(\frac{\pi n x}{2}\right) dx \right]$$

$$= \frac{x}{\pi n} \sin\left(\frac{\pi n x}{2}\right) + \frac{2}{\pi^2 n^2} \cos\left(\frac{\pi n x}{2}\right) \Big|_0^2$$

$$= \frac{2}{\pi n} \sin(\pi n) + \frac{2}{\pi^2 n^2} \cos(\pi n) - \left(0 + \frac{2}{\pi^2 n^2}\right)$$

$$= \frac{2}{\pi^2 n^2} (-1)^n - \frac{2}{\pi^2 n^2}$$

$$= \frac{2}{\pi^2 n^2} [(-1)^n - 1]$$

$$u = x \\ du = dx$$

$$v = \frac{2}{\pi n} \sin\left(\frac{\pi n x}{2}\right) \\ dv = \cos\left(\frac{\pi n x}{2}\right) dx$$

Input answers here:

(a) Exponential Form: $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi nx/2}$ where $d_n = \begin{cases} 1 & n = 0 \\ \frac{2}{\pi^2 n^2} [(-1)^n - 1] & n \neq 0 \end{cases}$

$$a_0 = 1$$

(b) Cosine & Sine Form: $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{\pi nx}{2}\right) + b_n \sin\left(\frac{\pi nx}{2}\right)$ where $a_n = 2d_n = \frac{4}{\pi^2 n^2} [(-1)^n - 1]$

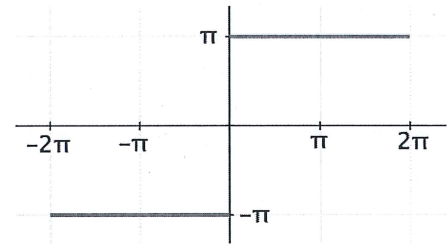
Since $f(x)$ is even $\rightarrow b_n = 0$

Problem 2: For the function depicted below, determine the Fourier Series in:

(a) Exponential Form: $e^{i\pi nx/L}$

(b) Cosine & Sine Form: $\cos\left(\frac{\pi nx}{L}\right)$ & $\sin\left(\frac{\pi nx}{L}\right)$

Write your answers in the same format as in **Problem 1**.



(assume periodic extension along the x-axis)

ODD and $L = 2\pi$

$$d_0 = 0$$

$$\begin{aligned} d_n &= \frac{-i}{2\pi} \int_0^{2\pi} \pi \sin\left(\frac{1}{2}nx\right) dx \\ &= \frac{i}{2} \left(-\frac{2}{n} \cos\left(\frac{1}{2}nx\right) \right) \Big|_0^{2\pi} \\ &= \frac{i}{n} \left(\cos(\pi n) - \cos(0) \right) \\ &= \frac{i}{n} \left[(-1)^n - 1 \right] \end{aligned}$$

Input answers here:

(a) Exponential Form: $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\frac{1}{2}nx}$

where $d_n = \begin{cases} 0 & n = 0 \\ \frac{i}{n} [(-1)^n - 1] & n \neq 0 \end{cases}$

(b) Cosine & Sine Form: $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n}{2}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n}{2}x\right)$

where $a_0 = 0$ ← Since f is odd
 $a_n = 0$
 $b_n = 2id_n = -\frac{2}{n} [(-1)^n - 1]$

Problem 3: SETUP the integral (or sum of integrals) needed to compute the Fourier coefficients of $f(x)$ (i.e. d_0 & d_n). Fully complete d_0 & d_n up to the point where you would begin to find the anti-derivative. Assume $f(x)$ periodic extends along the x-axis.

$$d_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^0 (-x-\pi) dx + \int_0^{\pi/2} x dx + \int_{\pi/2}^{\pi} (-x+\pi) dx \right]$$

$$d_n = \frac{1}{2\pi} \left[\int_{-\pi}^0 (-x-\pi) e^{-inx} dx + \int_0^{\pi/2} x e^{-inx} dx + \int_{\pi/2}^{\pi} (-x+\pi) e^{-inx} dx \right]$$

